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Aim: To study Project Scheduling using Gantt Chart.

Theory:

What is a Gantt Chart?

A Gantt chart is a type of bar chart that illustrates a project schedule. It shows the start and finish dates of the various elements of a project, making it easy to see what needs to be done, when, and by whom.

Key Components

- **Tasks/Activities:** Each row represents a specific task or activity.
- **Timeline:** The horizontal axis shows time (days, weeks, months).
- **Bars:** Horizontal bars represent the duration of each task, stretching from the start date to the end date.
- **Dependencies:** Lines or arrows may connect tasks to show dependencies (i.e., Task B cannot start until Task A is finished).
- **Milestones:** Special symbols or markers indicate significant events or deadlines.

How Gantt Charts Work

- Tasks are listed vertically, and time is shown horizontally.
- Each task's bar length corresponds to its duration.
- Overlapping bars show which tasks run concurrently.
- Dependencies help identify the critical path (the sequence of tasks that determines the minimum project duration).

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Benefits

- **Clarity:** Provides a clear visual overview of the project timeline.
- **Tracking Progress:** Easily see which tasks are on track, delayed, or completed.
- **Resource Management:** Assign resources and avoid overloading team members.
- **Identifying Bottlenecks:** Spot dependencies and potential delays early.

Limitations

- Can become complex for large projects with many tasks.
- Requires regular updates to remain accurate.
- May not show task details or resource allocation unless enhanced.

Video Downloader App:

1. Tasks/Activities

Each of these is a row/bar in the Gantt chart:

- Requirements Gatheringcomplete
- Identify user needsto do
- Develop minimal viable prototype idea for the appto do
- Create wireframesto do
- Create prototype wireframescomplete
- Develop prototype featuresto do
- UI/UX Designto do
- Frontend completein progress
- Backend Developmentto do

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- Testing & QAto do
- App Deliveryto do
- Maintenance and supportto do

2. Timeline (Start and Due Dates)

- Each task has a start and/or due date, which determines its position and length on the Gantt chart.
- Example: "Requirements Gathering" runs from 02/22/2026 to 02/24/2026, "App Delivery" is scheduled for 03/15/2026 to 03/16/2026.

3. Milestones

- "Frontend complete" is marked as a milestone, representing a key project checkpoint.

4. Task Status

- Tasks have statuses like "to do", "in progress", or "complete", which can be visualized on the Gantt chart for tracking progress.

5. Task Descriptions

- Each task can have a description, providing more context (e.g., "UI/UX Design" includes wireframes and mockups).

6. Dependencies (Manual/Comments)

- Some dependencies are noted in comments, such as "add this after ui ux design" for the "Frontend complete" milestone.
- You can use these to set up dependencies in your Gantt view for accurate sequencing.

7. Progress Tracking

- Completed tasks like "Requirements Gathering" and "Create prototype wireframes" show project advancement.

8. Grouping/Phases

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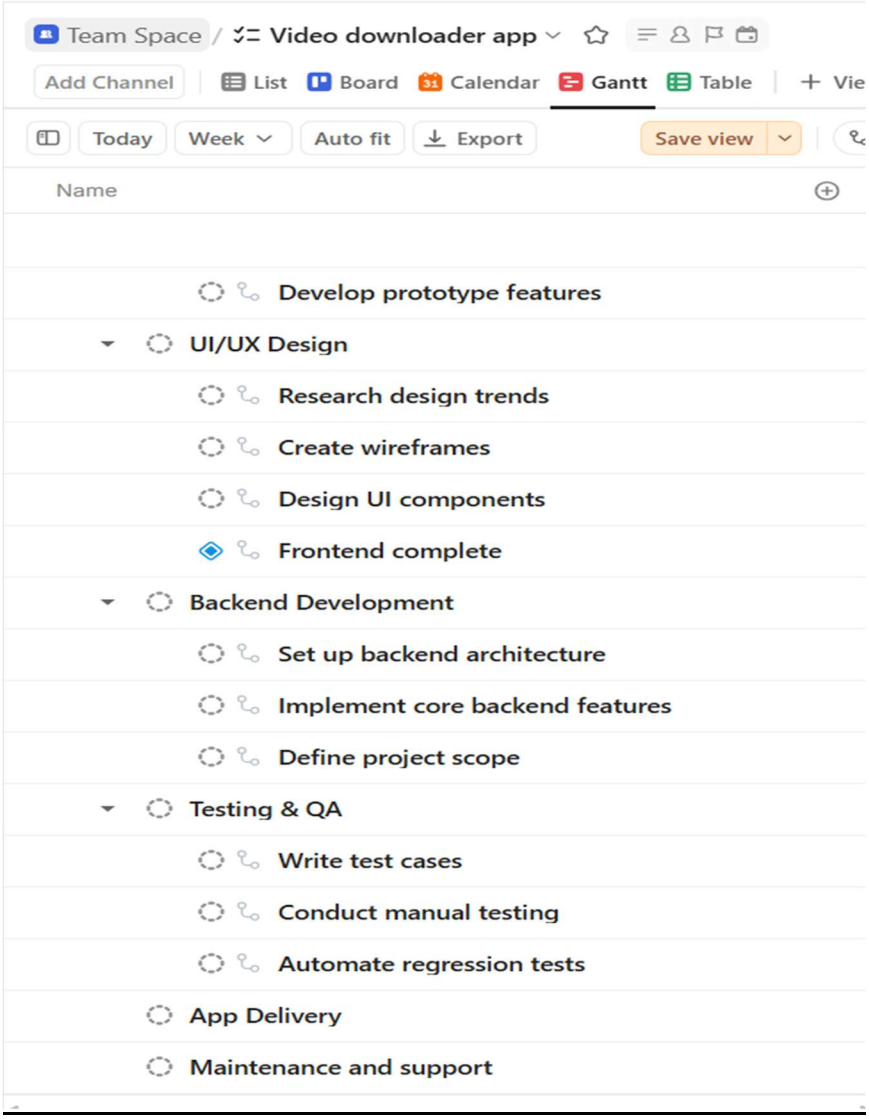
- Tasks are naturally grouped by project phase: requirements, design, development, testing, delivery, and support.



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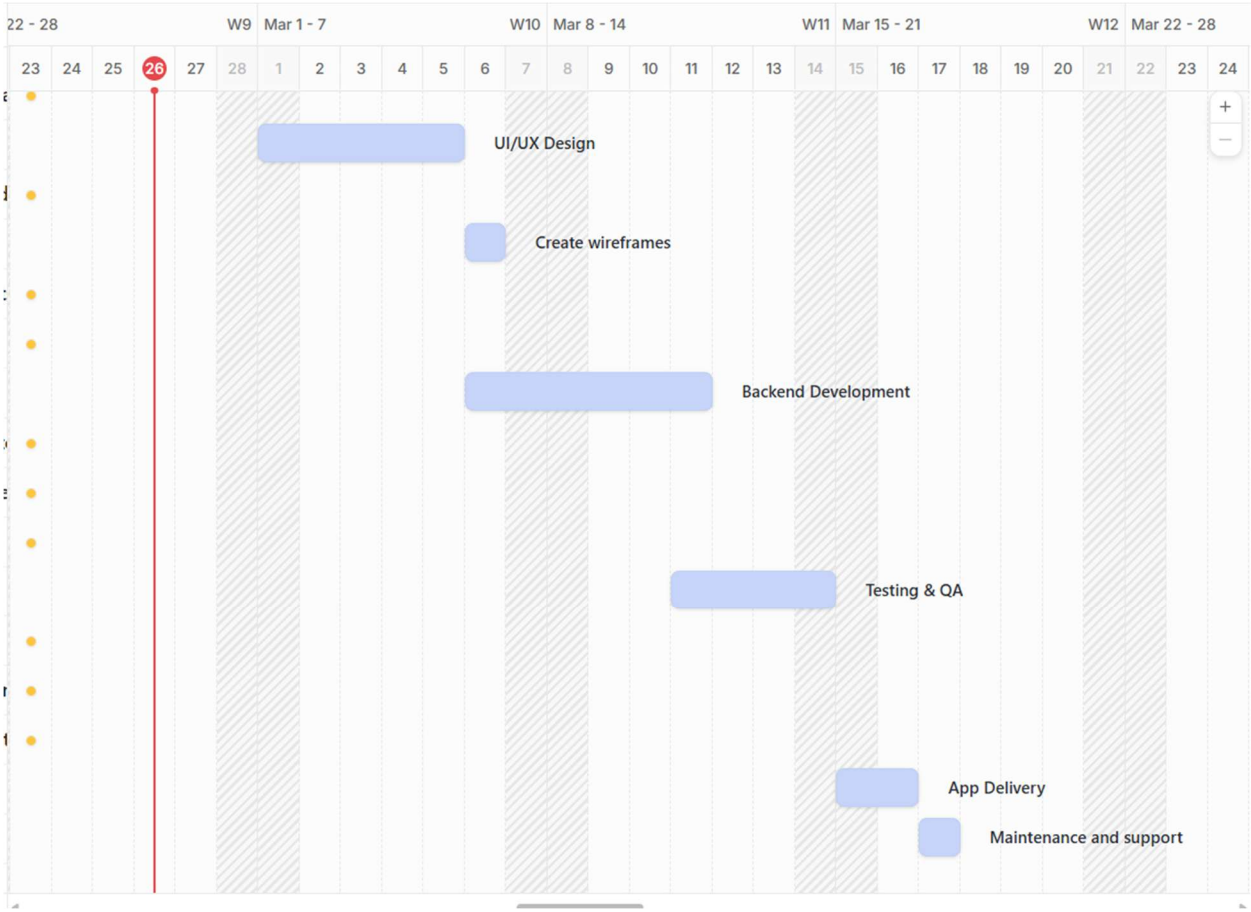

<u>Conclusion:</u>

Hence we have studied and implemented gantt chart in clickup.